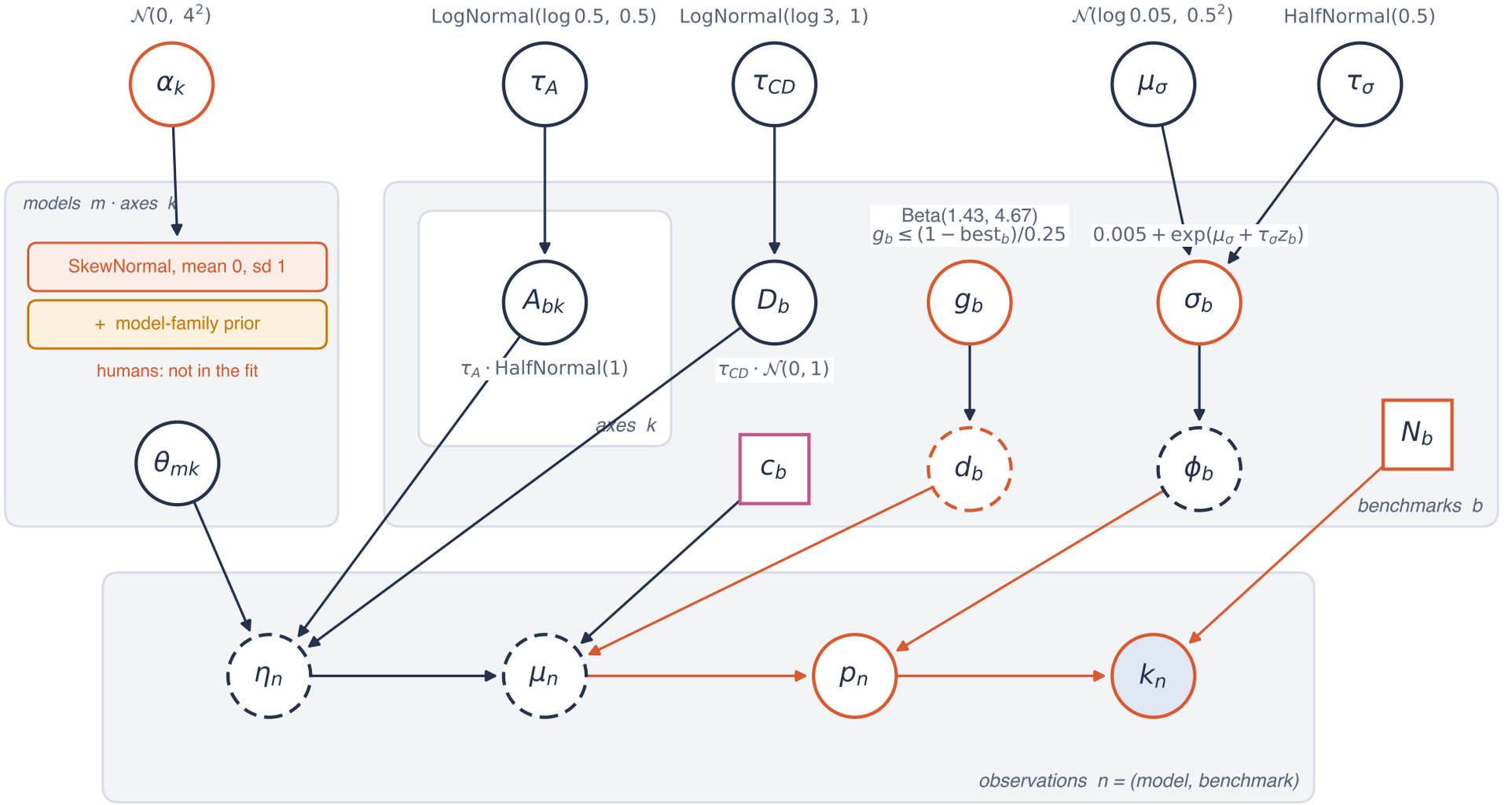
Beta-MIRT model (K = 4), flagship on 2026-11-03



$$\eta_n = \sum_k A_{b_n k}\,\theta_{m_n k} - D_{b_n}$$

$$\mu_n = c_{b_n} + (d_{b_n} - c_{b_n})\,\text{logistic}(\eta_n), \qquad d_b = 1 - 0.25\,g_b$$

$$p_n \sim \text{Beta}(\mu_n\phi_{b_n},\ (1-\mu_n)\phi_{b_n}), \qquad \phi_b = \tfrac{1}{4\sigma_b^2} - 1$$

$$k_n \mid p_n \sim \text{Binomial}(N_{b_n},\ p_n) \quad \text{so} \quad k_n \sim \text{BetaBinomial}(N_{b_n},\ \mu_n\phi_{b_n},\ (1-\mu_n)\phi_{b_n})$$

82 / 107 benchmarks. The other 25 have no question count: there the score y_n itself is drawn, $y_n \sim \text{Beta}(\mu_n\phi_{b_n},\ (1-\mu_n)\phi_{b_n})$, zeros set to 0.005.

random · computed from its parents · observed · fixed, read from a file · new or changed since docs/prior_graph.pdf

Symbols

Indices

m	a model (a test-taker), e.g. GPT-5 at medium effort
b	a benchmark, e.g. SimpleBench
k	an axis of ability, k = 1 ... 4
n	one observation: the score of model m_n on benchmark b_n

Data

N_b	number of questions on benchmark b (from the item-count file)	new
k_n	number of questions model m_n answered right on b_n: score x N_b	new
y_n	the score itself, in [0, 1]; used only where N_b is unknown (25 / 107)	
c_b	chance floor: the score of random guessing (0.25 on 4-choice questions)	
best_b	highest score anyone, model or human, reached on b	

Abilities and benchmarks

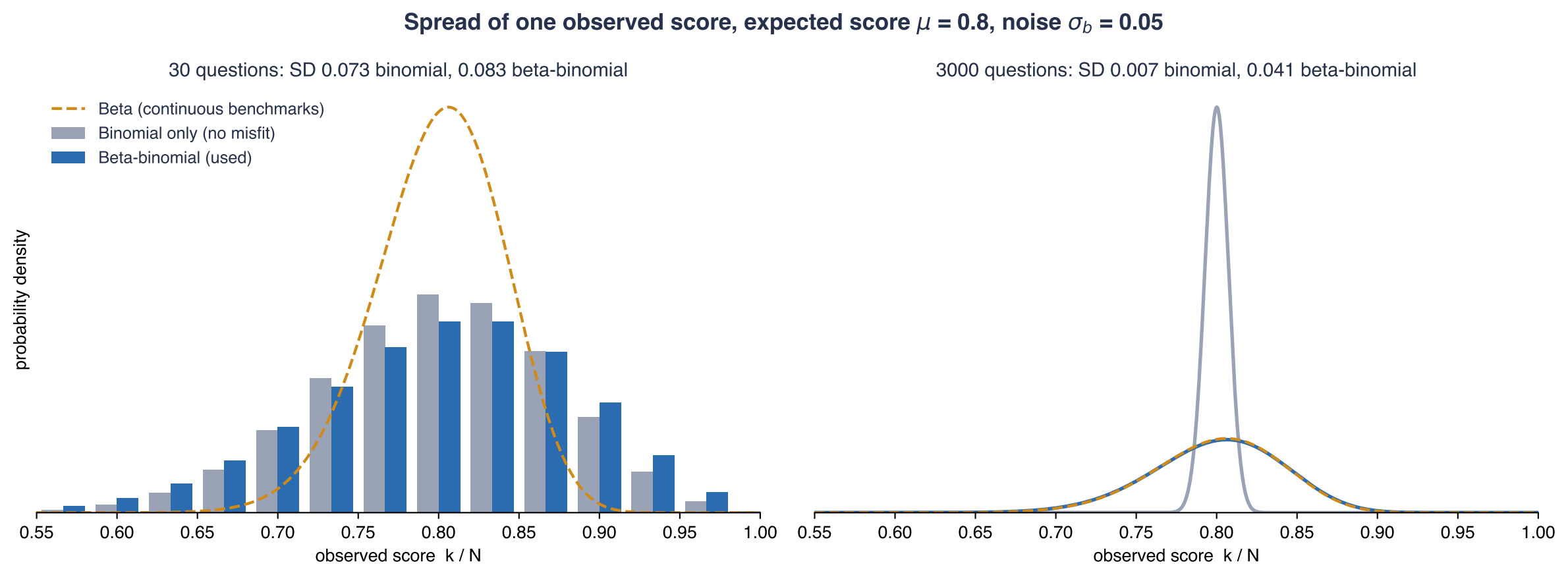
θ_{mk}	ability of model m on axis k	
α_k	skewness of axis k's abilities: 0 is a symmetric bell, < 0 a long left tail	new
A_{bk}	loading: how strongly benchmark b's score rises with axis k (>= 0)	
D_b	difficulty of benchmark b	
$\tau_A,\ \tau_{CD}$	typical size of the loadings, and of the difficulties	

From ability to expected score

η_n	ability minus difficulty for this pair, on the logit scale	
$\text{logistic}(x)$	$1/(1+e^{-x})$: turns a logit into a share in (0, 1)	
g_b	how far the ceiling sits below 1, as a share of the largest gap allowed (0.25)	new
d_b	ceiling: the best expected score a perfect model gets, $1 - 0.25\,g_b$	new
μ_n	expected score of model m_n on b_n: between c_b and d_b	

Noise

σ_b	noise of benchmark b: the score's SD when $\mu = 0.5$ and N_b is large; floor 0.005	new
z_b	a standard normal draw, one per benchmark	
$\mu_\sigma,\ \tau_\sigma$	typical log-noise across benchmarks, and its spread	
ϕ_b	precision: $1/(4\sigma_b^2) - 1$; large ϕ means little scatter	
p_n	the success rate model m_n really has on b_n's questions: μ_n plus misfit	new


Spread of one observed score, expected score μ = 0.8, noise σb = 0.05
30 questions: SD 0.073 binomial, 0.083 beta-binomial
3000 questions: SD 0.007 binomial, 0.041 beta-binomial
Beta (continuous benchmarks)
Binomial only (no misfit)
Beta-binomial (used)
probability density
0.55
0.60
0.65
0.70
0.75
0.80
0.85
0.90
0.95
1.00
observed score k / N